29/9/26

Activity - 1

Another Example:-

Target m/c code

Source program

Position = initial + rate * 60

↓

Lexical Analyzer

i = j + k * 60

j = id 1

J = id 2

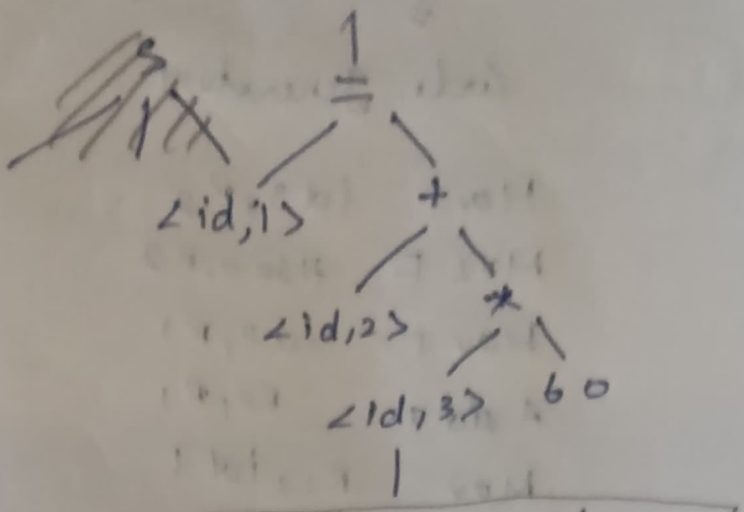
K = id 3

< id, 1 > < = > < id, 2 > < + > < id, 3 > < * > < 6

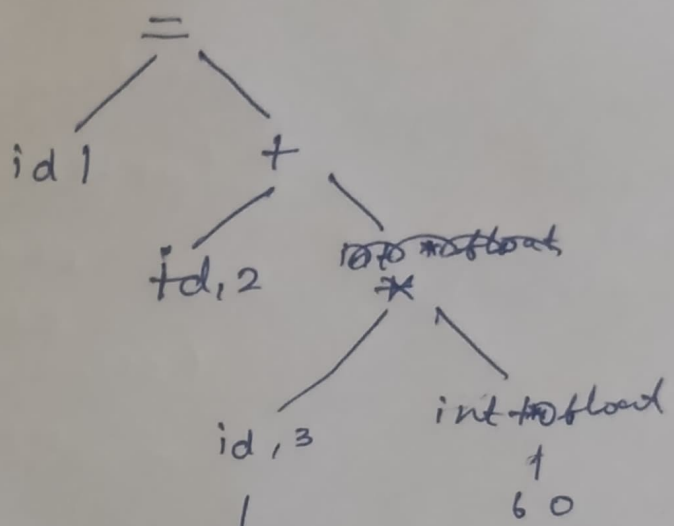
$id_1 = id_2 + id_3 * 60$

↓

Syntax Analyser



Semantic Analyzer.



intermediate code generator

$t_1 = int\ to\ float\ 60$

$t_2 = id3 * t1$

$t_3 = id2 + t2$

$id1 = t3$

Code optimizer

$t_1 = id3 * 60$

$t_2 = id2 + t1$

$id1 = t2$

↓

Code Generator

```
MovF   id3, R2
MVLF   #60.0, R2
MovF   id2, R1
AddF   R2, R1
Mov    R1, id1
```